

Southface Eco Office Achieves 80% Water Savings

by Sydney G. Roberts
and Paul Morgan

For the Atlanta metro area, rainfall has become both a blessing and a curse. While the area receives abundant precipitation, with over 50 inches of rain falling over an average of 115 days each year, Atlanta's aging storm water and sewage system is currently incapable of handling the downfall from even moderately heavy storms. This means that even though nature supplies sufficient water to the region for today's demands, the Atlanta Regional Commission predicts that by 2025 Metro Atlanta will have outgrown its supply of potable water. Further, water and sewer rates have increased nearly 100% in the past five years. It is therefore critical to act now to reduce both indoor and outdoor use of potable water.

The Southface Eco Office, a 10,000-square foot office building in the final stages of construction in downtown Atlanta, incorporates a variety of water-conserving and water-harvesting strategies that will reduce potable water use by 80%. Although the Eco Office is technically a commercial building, these strategies are appropriate for many residential projects as well. And although this one building represents a mere drop in the bucket in terms of the pending water crisis, hundreds of thousands of design and construction professionals are expected to read about, or even visit, this building. What they learn there will lead to reduced consumption—residential as well as commercial—for the region as a whole.

To help reduce water consumption, the Eco Office has low-flow or no-flow fixtures and Energy Star



The Southface Eco Office in Atlanta, Georgia, will demonstrate a wide range of water-saving technologies.

appliances throughout. The first-floor restrooms, serving groups of up to 100, are equipped with Caroma Caravelle dual-flush toilets and Mansfield QuantumOne pressure assist toilets. The Caravelle offers two flushing options—a full flush, which uses the standard 1.6 gallons per flush, and a half flush, which uses only 0.8 gallons. Although these toilets are constructed with traditional plumbing, the option to use the half flush is projected to save 20,000 gallons per year. The QuantumOne uses a pressurized tank to flush with only 1 gallon—a savings of 38% over a standard toilet.

Restrooms on the third floor, or office level, will serve 20 people. The restrooms' fixtures include two Clivus Multrum foam flush composting toilets (see Figure 1 and photo, p. 17). The foam flush toilet looks and functions

like a conventional toilet. However, it uses much less water per flush and produces no sewage. Instead of relying solely on water to flush, the toilet uses a mixture of biocompatible soap and water to carry waste to the composting system below. Each flush uses 6 ounces of water—only 3% of the amount used by a conventional toilet. We estimate that this system will save more than 20,500 gallons of water per year compared to a pair of toilets in a typical office building. While the foam flush toilets in this building are sized for an office, residential-sized units are also available. These toilets require little maintenance and create no odor.

The urinals in the Eco Office are either Sloan Waterfree or Clivus Multrum foam-flush toilets. Both are essentially waterless urinals; however, the Clivus Multrum uses a mixture of

biocompatible soap and water to carry toilet waste to a composting system. There is no water supply or flush valve in the Sloan Waterfree urinal, only a replaceable and biodegradable cartridge. This cartridge contains a biocompatible fluid sealant that acts as a vapor lock and prevents any odors from wafting up to the urinal user. Urine flows through the cartridge, which traps any sediment along with the odors, and down into the drainpipe. According to Sloan, the cartridge can be used 7,000 times before it needs to be replaced. These waterless urinals are estimated to save approximately 25,000 gallons of water per year compared to conventional fixtures. Several manufacturers have begun to market waterless and low-flow urinals directly to residential customers.

All lavatories for the public and staff are supplied with cold water only. This has reduced the initial cost of installation and will save energy. Because the Eco Office requires relatively little hot water, only the janitor closets and the staff kitchen are equipped with small electric Chronomite Instant-Flow water heaters.

Sloan Solis low-flow automatic faucets are installed in all of the building's lavatories. These faucets are equipped with solar-powered, or ambient-light-powered, electronic motion sensors that turn the water on and off. The faucet uses a Power Plant storage cell to turn ambient light into electrical energy. The water flow averages 0.5 gallons per minute compared to the 2.5 gallons per minute of a conventional faucet. This low-flow feature will save more than 47,000 gallons of water per year. In the staff kitchen and catering staging areas, Whirlpool Energy Star-qualified dishwashers will consume 44% less water than conventional units.

In addition to minimizing the water used by the fixtures in the Eco Office, we have also designed our plumbing system so that no potable water will be used to flush toilets. All of the water used for this purpose will be rainwater

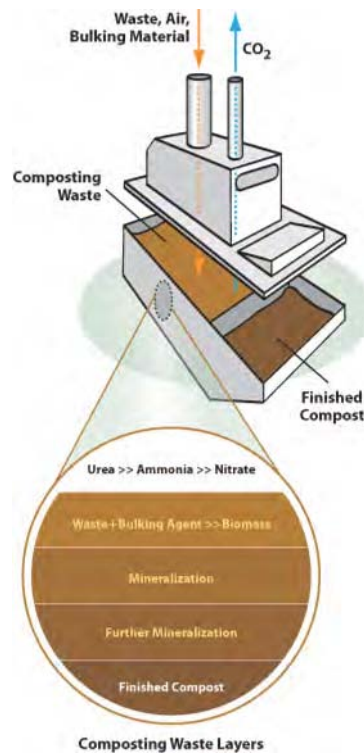


Figure 1. This composting toilet uses bio-compatible soap and water to flush waste to a composting system.



Composting toilets by Clivus Multrum look like standard fixtures, but require just 6 ounces of water per flush.

collected in rooftop and underground cisterns. The new 2,040- cubic foot Atlantis Water Management Systems RainTank cistern will store over

14,500 gallons of rainwater for flushing toilets and cooling mechanical heat exchangers in the Eco Office. The RainHarvest Company has installed similar cisterns in other commercial and residential properties in Metro Atlanta. The RainTank, manufactured from 100% postconsumer plastic, is composed of individual panels, which snap together to create permeable cubes, forming the skeleton of the underground cistern. Because the system is composed of individual cubes, smaller systems can be installed for residential use. The cistern is wrapped with a 30-mil EPDM—a rubber material—membrane to contain the rainwater. A Deep Root physical root barrier protects this membrane from thirsty tree roots without the use of herbicides. The Deep Root product has a high postconsumer recycled content as well.

In most residences, landscaping accounts for almost 50% of the water used—and usually all of that water is potable. The Eco Office site has been landscaped to include a rain garden and native, drought-resistant plants. This landscaping will provide four-season interest with little watering or other maintenance. The site has been graded to reduce runoff to the storm water system.

Harvested rainwater will be used to supply drip irrigation to establish the site's drought-resistant landscaping. Weathermatic controls include the Smartline system with an SLW Series On-Site Weather Station. This system will calculate real time evapotranspiration, and can be zoned to respond to different plant types, soil types, shade levels, and wind conditions.

Rainwater from the roof of the Eco Office and from part of the original office building will flow into the underground cistern via 6-inch polyvinyl chloride drain lines. Storm water runoff from the northern and eastern parts of the site will be collected in a system of swales and will percolate in through the top of the tank. A layer of filter fabric below a healthy and dense turf will prevent

any sediment from entering the tank. Capturing the majority of the storm water will prevent water from running off the site. Normally this water would enter the storm sewers, burdening the municipal waste system.



The Atlantis Water Management System RainTank cistern, shown here during installation at Southface, is customizable for any sized outdoor space and is made of recycled plastic.

Part of the Eco Office roof will be a green roof, which means that it will be covered with a waterproof membrane, a layer of a growing medium, and plants. The green roof will capture and filter rainwater, provide insulation, and reduce the heat island effect. Water filtered through the green roof will be monitored for water quality, just to see if it meets the standard for potable water without requiring sterilization. All water from the roof will be captured in the cistern system.

True to Southface's mission of research and education, water use at every fixture in the Eco Office will be monitored by a Lucid Design Group building monitoring system known as the Building Dashboard. Anyone can access water and energy consumption data for the building online, using this

system. Both real-time and cumulative data will be available.

Southface is applying for the U.S. Green Building Council's LEED-NC Platinum certification for the Eco Office. Southface expects to receive

high points in the water efficiency and rainwater reclamation categories. With this building, Southface is also setting a new standard in water conservation for the Atlanta community. The purchase and use of potable water on campus will decrease by approximately 80%, a tangible savings, which should return the investment in the water-efficient fixtures and water-harvesting products within seven years. The intangible benefits of these water savings are manifold. Atlanta's limited water supply will not be heavily taxed by this development. The Atlanta sewer system, already overburdened, will no longer have to handle storm water from the site. The Southface Eco Office demonstrates successful water-conserving and

water-harvesting strategies that should lead residential and commercial development toward a more sustainable future.



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For more information:

Visit the Southface Web site at www.southface.org.